

Input Performance

KLM, Fitts' Law, Pointing Interaction Techniques

Input Performance Models

- You're designing an interface and would like to:
 - choose between candidate designs without building them
 - estimate performance with your new design
- Solution: use a model of how people use input devices and interfaces to predict time, error, fatigue, learning, etc.
 - models most often focus on time and error (easiest to measure)

"MM/DD/YYYY"

A single rectangular text input field with a thin border, set against a light gray background.A date input field consisting of three adjacent boxes. The first box contains the text 'May' with a small downward arrow on its right side. The second box contains the text '22' with a small downward arrow on its right side. The third box contains the text '1997' with a small downward arrow on its right side. The entire field is set against a light gray background.A date input field consisting of three adjacent empty rectangular boxes. Below each box is a label: 'Month' under the first, 'Day' under the second, and 'Year' under the third. The entire field is set against a light gray background.

Keystroke Level Model (KLM)

- Describe each task with a sequence of operators
- Sum up times to estimate how long the task takes
- Operator types
 - K** Keystroke = 0.08 – 1.2s (based on expertise, type of string)
 - P** Pointing = 0.8s to 1.5s with an average of 1.10s
 - B** Button press on mouse = 0.1s
 - H** Hand move from mouse to/from keyboard = 0.4s
 - M** Mental preparation = 1.2s
- Great online resource for KLM (Kieras, 1993):
 - <http://web.eecs.umich.edu/~kieras/docs/GOMS/KLM.pdf>

KLM Operators



Code	Operation		Time
K	Key press and release (keyboard)	Best Typist (135 wpm)	0.08 seconds
		Good Typist (90 wpm)	0.12 seconds
		Poor Typist (40 wpm)	0.28 seconds
		Average Skilled Typist (55 wpm)	0.20 seconds
		Average Non-secretary Typist (40 wpm)	0.28 seconds
		Typing Random Letters	0.50 seconds
		Typing Complex Codes	0.75 seconds
		Worst Typist (unfamiliar with keyboard)	1.20 seconds
P	Point the mouse to an object on screen		1.10 seconds
B	Button press or release (mouse)		0.10 seconds
H	Hand from keyboard to mouse or vice versa		0.40 seconds
M	Mental preparation		1.20 seconds

KLM Example (Only Physical Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: **hand already on mouse**, **40 WPM typist**)

- One text field

“MM/DD/YYYY”



PB H KKKKKKKKKK = 4.6s

- Steps:

1. Move the mouse cursor to the box/edittext (P)
2. Click on the box/edittext (B)
3. Move the hand from the mouse to the keyboard (H)
4. Type 10 characters, including “/” (10K)

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

KLM Example (Only Physical Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: **hand already on mouse**, **40 WPM typist**)

- Three Dropdowns



PBPB PBPB PBPB = 7.2s

- Steps:
 1. Move the mouse cursor to the first dropdown (P)
 2. Click on the first dropdown list to see the items (B)
 3. Move the cursor to select an item (P)
 4. Select an item (B)
- Repeat Step 1-4 for the second and third dropdown lists

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

KLM Example (Only Physical Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: **hand already on mouse**, **40 WPM typist**)

- One text field

“MM/DD/YYYY”



PB H KKKKKKKKKKK = 4.6s

- Three Dropdowns



PBPB PBPB PBPB = 7.2s

- Three text fields



With tab: PB H KK K KK K KKKK= 4.6s

Without tab: PB H KK HPB H KK HPB H KKKK= 8.0s

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

Including Mental Operators (M)

- People need to think about something before doing it
 - identify when people have to stop and think: **M**
 - difference between actions using *cognitive conscious* and *cognitive unconscious*
- Insert an **M** operation when people have to:
 - initiate a task
 - retrieve a chunk from memory
 - find something on the display (e.g. point to something)
 - think of a task parameter
 - verify that a specification/action is correct (e.g. display changes)

KLM Example (Including Mental Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: hand already on mouse, 40 WPM typist)

- One text field

"MM/DD/YYYY"



MPB H KKKKKKKKKKK = 5.8s

- Three Dropdowns



MPBMPB PBMPB PBMPB = 12s

- Three text fields



With tab: MPB H KK K KK K KKKK= 5.8s

Without tab: MPB H KK HPB H KK HPB H KKKK= 9.2s

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

KLM Critique

Benefits?

- Easy to model
- Can be done from just pictures or ideas

Drawbacks?

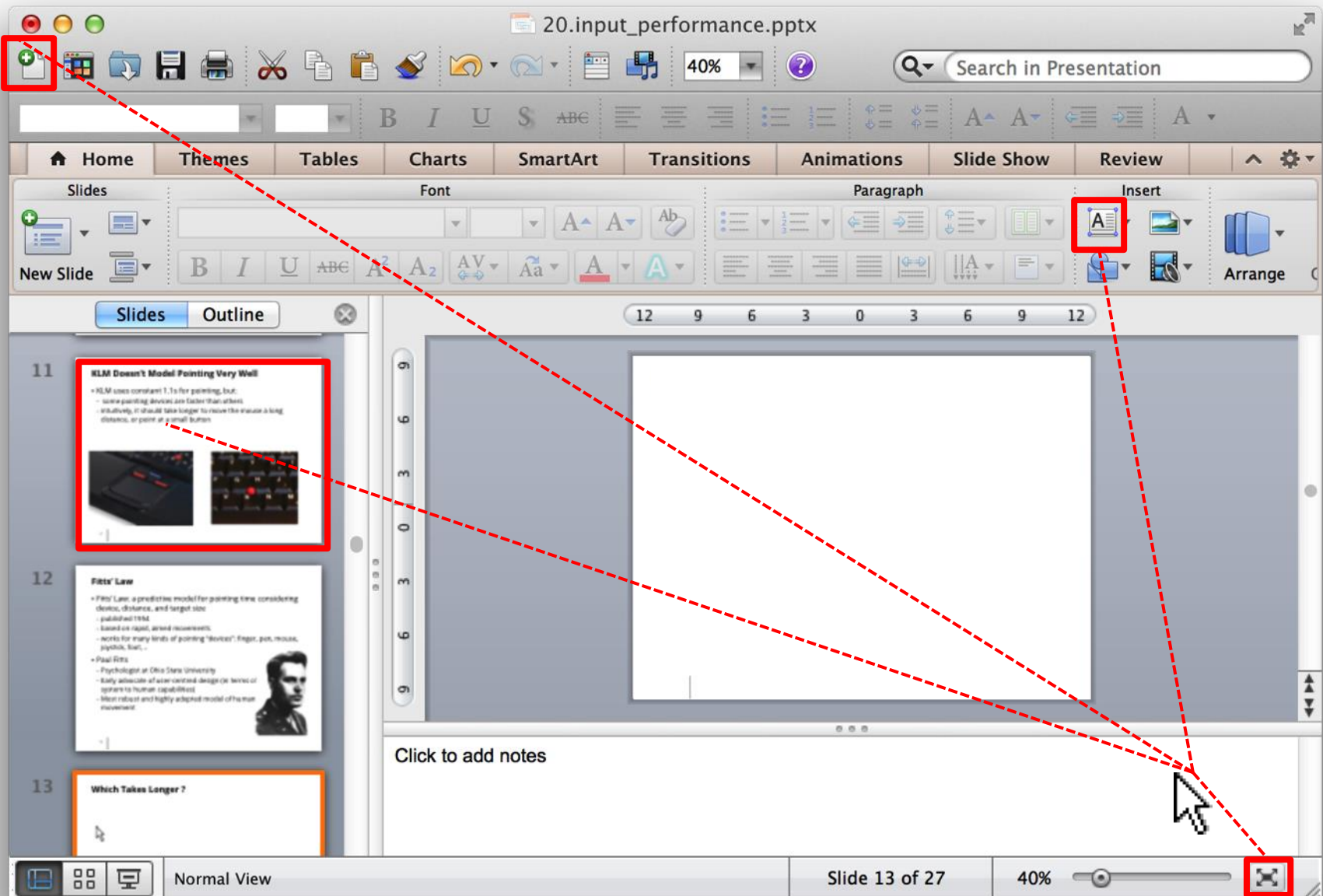
- Some time estimates are inherently variable
 - “P” for mouse? touch?
 - “P” for target size? distance?
- Out of date: what about soft keyboard?
- Doesn't model:
 - Errors
 - Learning time
 - etc.

KLM Doesn't Model Pointing Very Well

- KLM uses constant 1.1s for pointing, but:
 - Some pointing devices are faster than others
 - Intuitively, it should take longer to move the mouse a long distance, or point at a small button



Which Takes Longer?



Which Takes Longer?

Big targets are easier to hit than small targets
(Target Size)

+

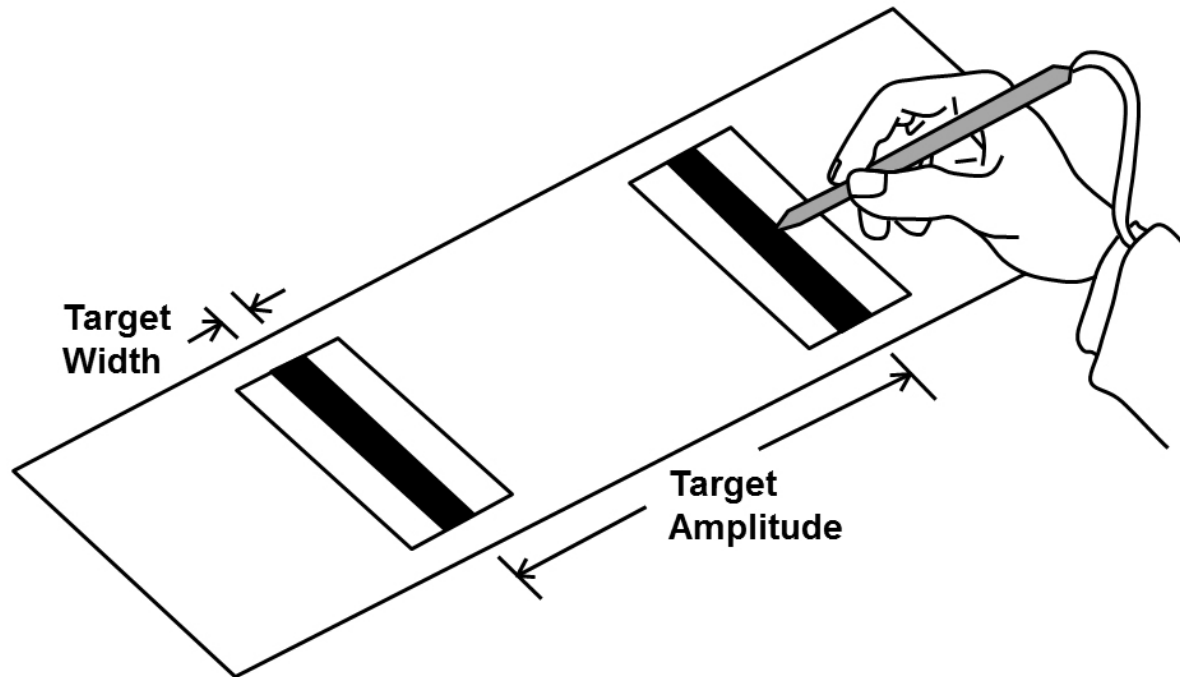
Close targets are easier to hit than far targets
(Target Distance/Amplitude)

Fitts' Law

- Fitts' Law: a predictive model for pointing time considering device, distance, and target size
 - published 1954
 - based on rapid, aimed movements
 - works for many kinds of pointing “devices”:
finger, pen, mouse, joystick, foot, ..
- Paul Fitts
 - Psychologist at Ohio State University
 - Early advocate of user-centred design
(in terms of matching system to human capabilities)



Paul Fitts' Experiment

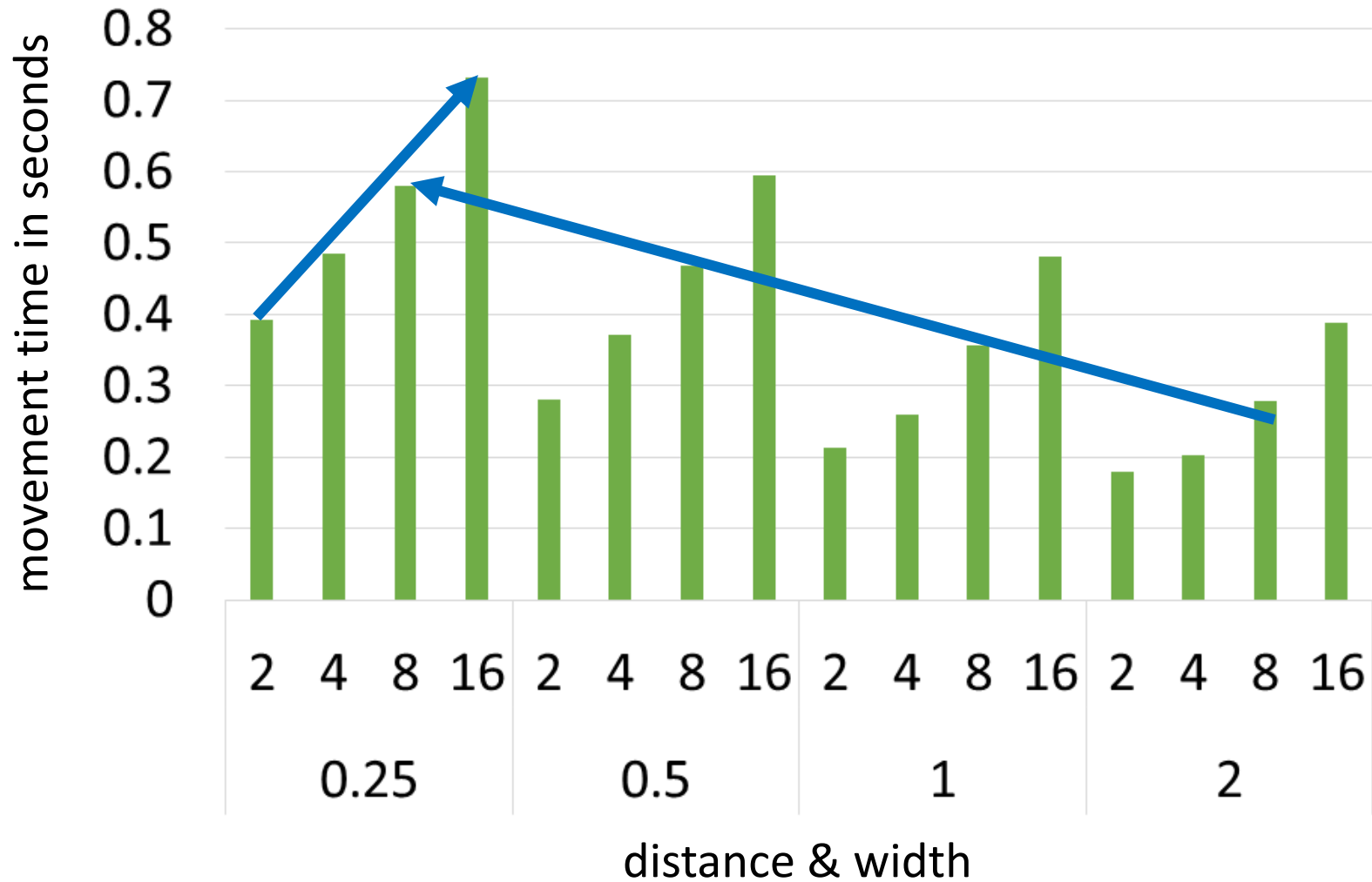


Four distances/Amplitudes: 2, 4, 8, 16 inch

Four widths: 0.25, 0.5, 1.0, 2.0 inch

- From: Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. *Journal of experimental psychology*, 47(6), 381.

Paul Fitts' Experiment



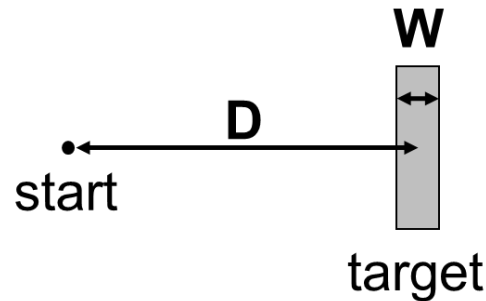
Distance vs. Size

- The larger the **distance**, the longer the **time**
- The smaller the **width** of the target, the longer the **time**
- So, a proportional relationship between movement time and distance and width:

$$MT \propto \frac{D}{W}$$

Distance vs. Size

The movement time (MT) to select a target is a function of the target's width (W) and distance (D). It depends on the input device.



$$MT = a + b \log_2 \left(1 + \frac{D}{W} \right)$$

MT: movement time

a & b: input device-dependent constants

D: distance to the target

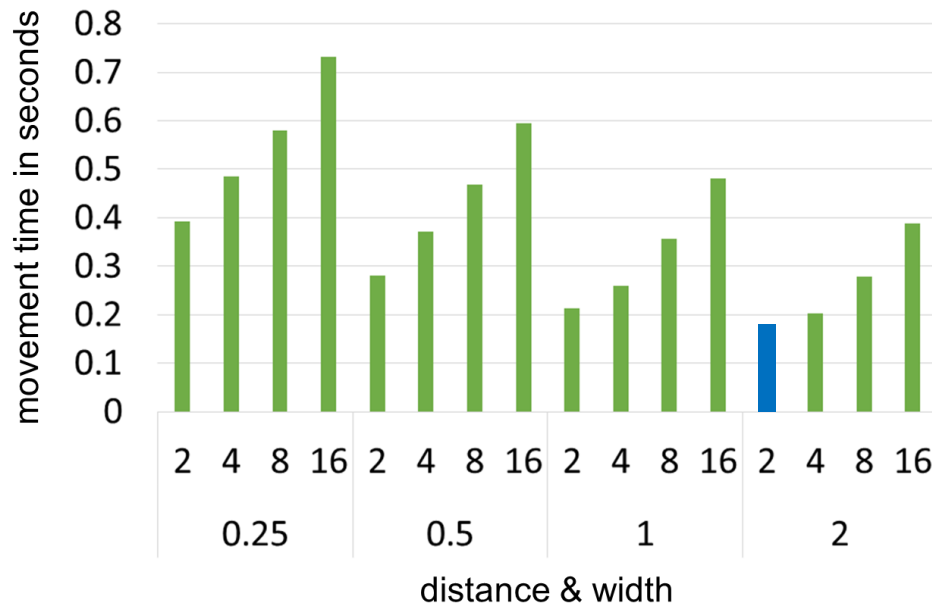
W: width of the target

Index of difficulty

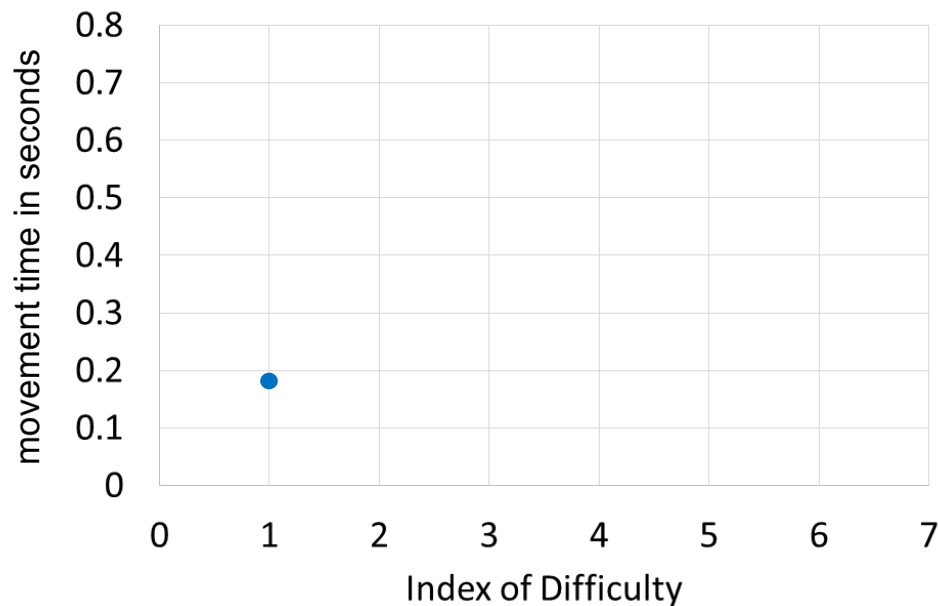
$$MT = a + b \underbrace{\log_2 \left(1 + \frac{D}{W} \right)}$$

- Index of Difficulty, ID = $\log_2 \left(1 + \frac{D}{W} \right)$
 - $MT = a + b \cdot ID$
 - ID how difficult a task is independent from the input device

- Units:
 - a is measured in seconds
 - b is measured in seconds per bit
 - Index of Difficulty (ID) is described in bits
 - a and b are constants determined through linear regression

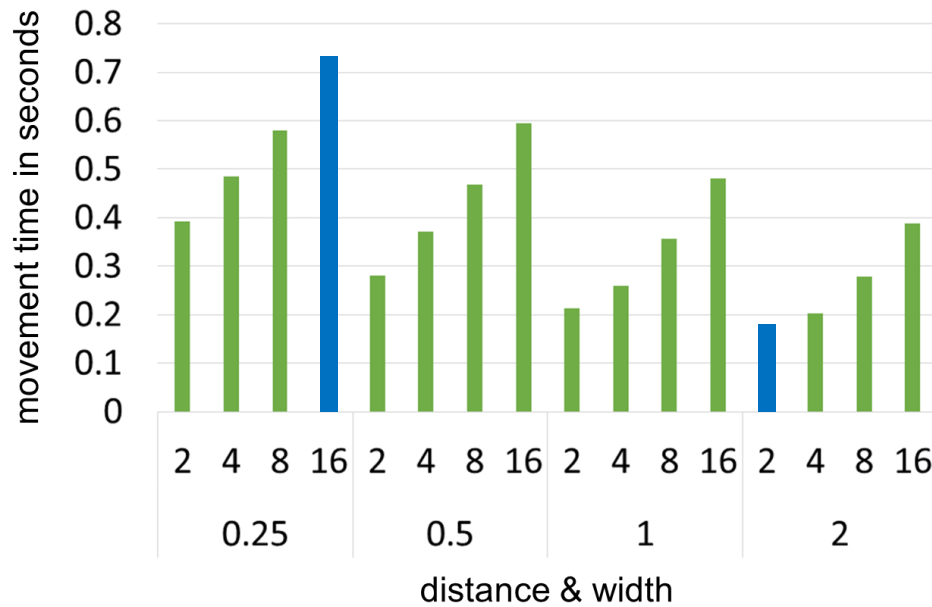


$$ID = \log_2 \left(1 + \frac{D}{W} \right)$$

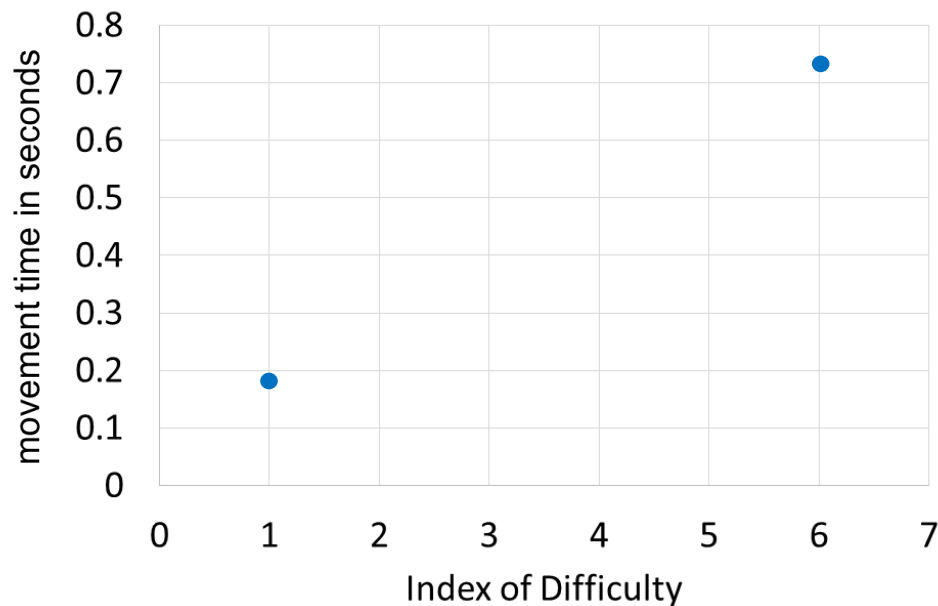


- $D=2, W=2$
- $ID = \log_2(1+1)=1$

ID Examples

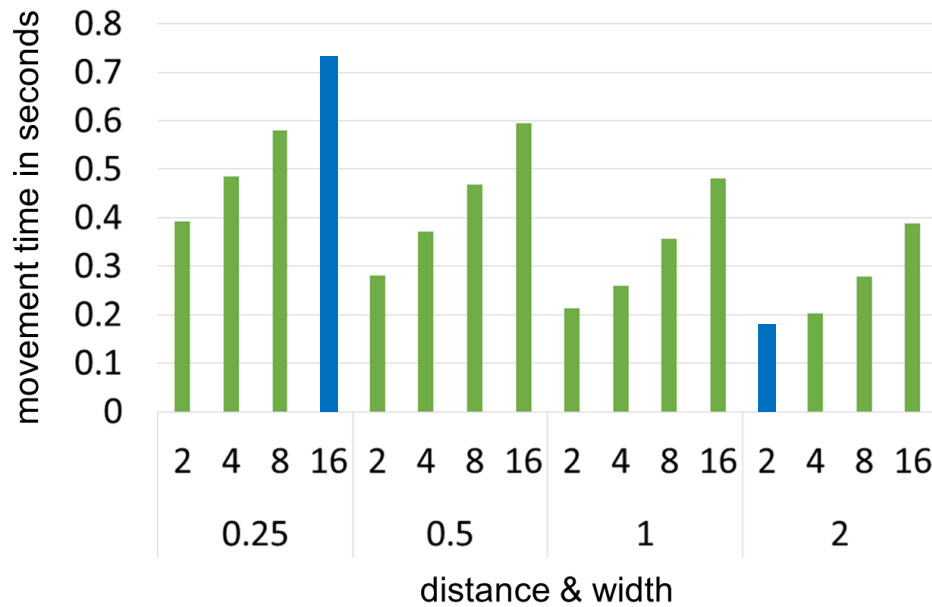


$$ID = \log_2 \left(1 + \frac{D}{W} \right)$$

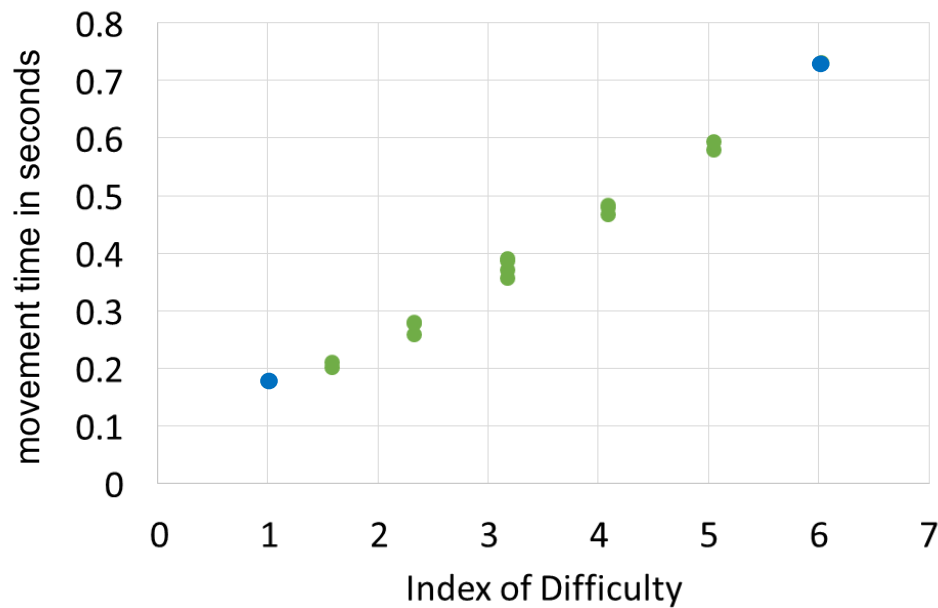


- D=16, W=0.25
- $ID = \log_2(1+64) = 6.02$

ID Examples

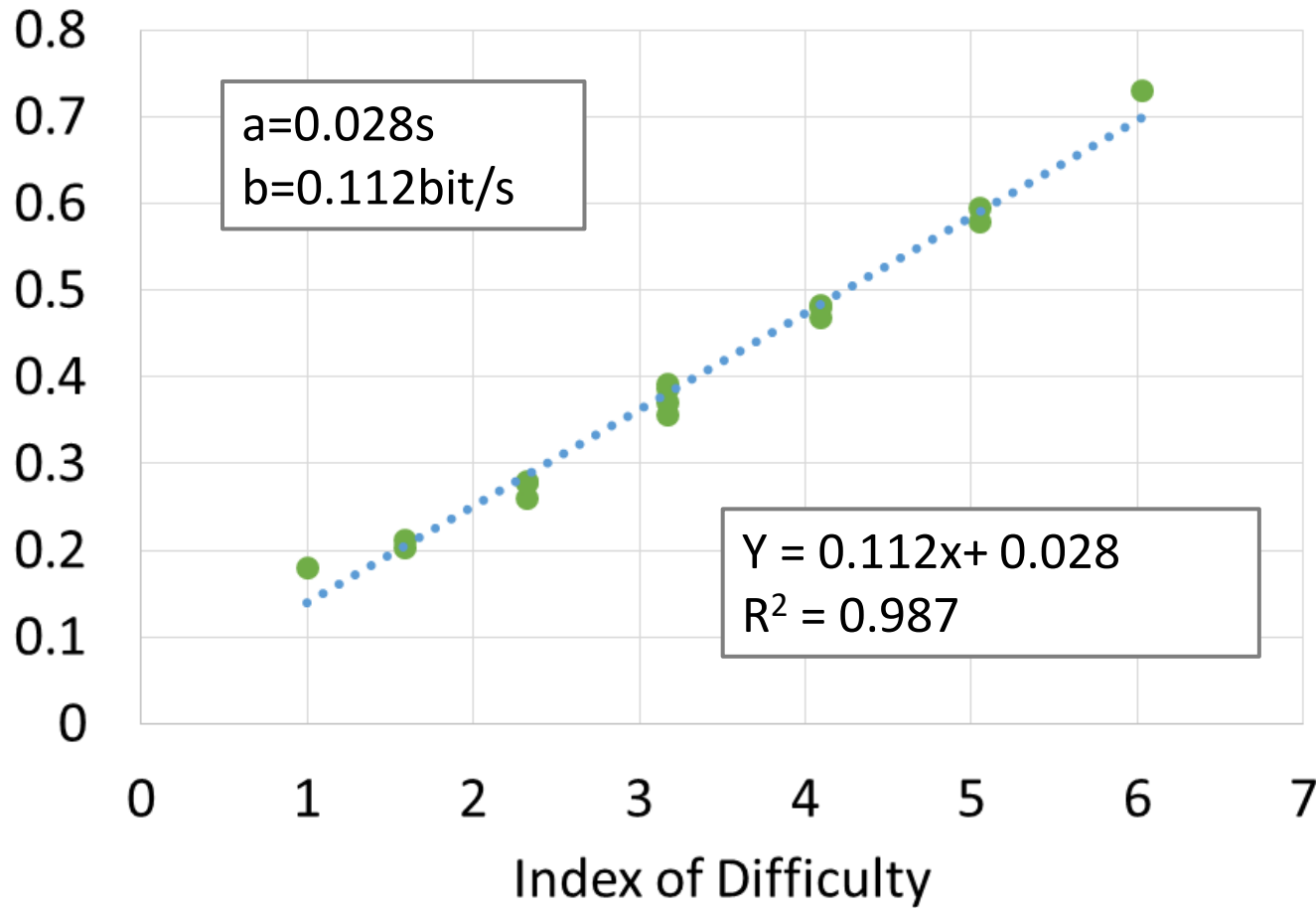


$$ID = \log_2 \left(1 + \frac{D}{W} \right)$$



ID Examples

Determining a and b



Predicting the movement time

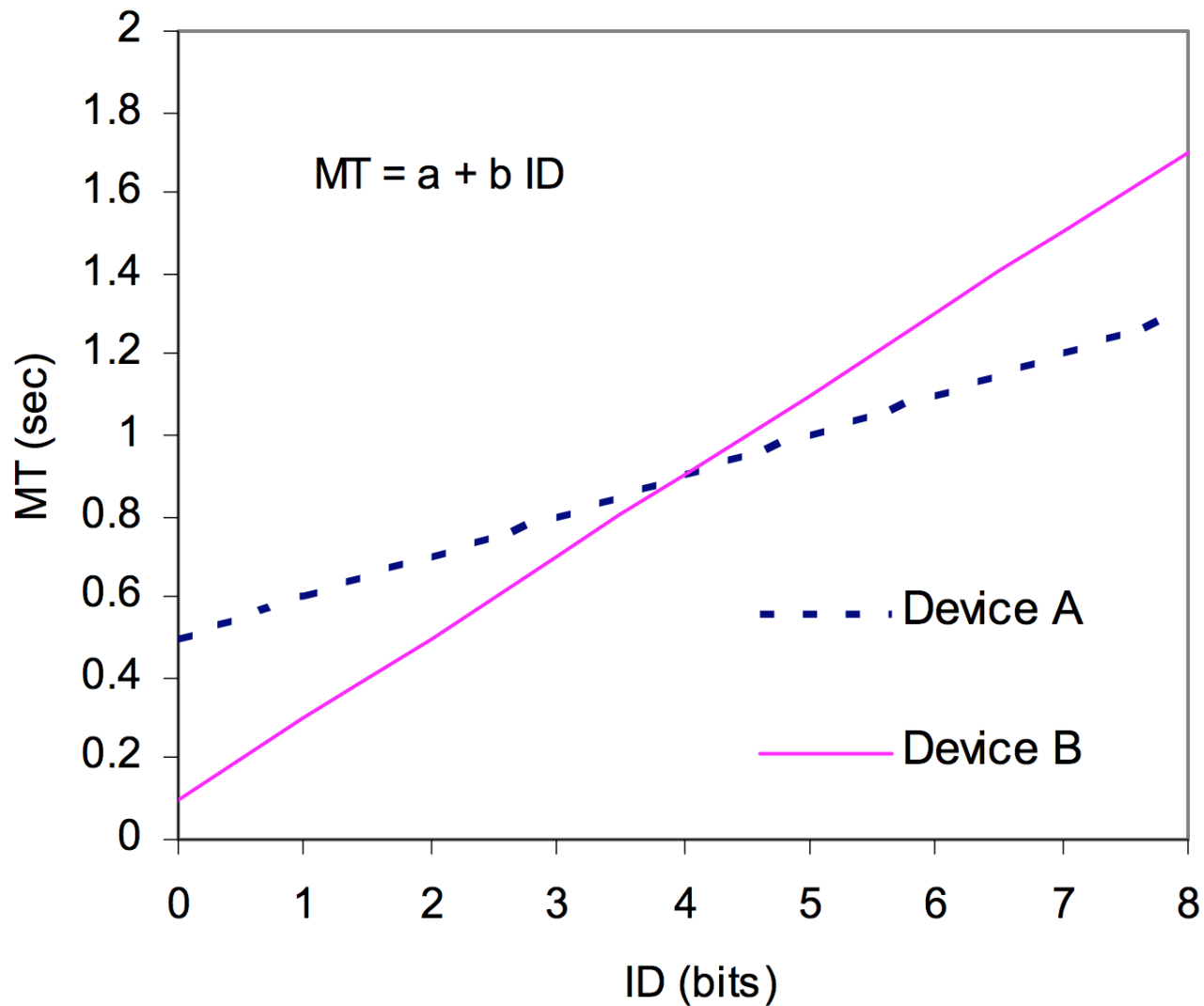
- $a=0.028s$
- $b=0.112s/bit$

$$MT = a + b \log_2 \left(1 + \frac{D}{W} \right)$$

- How long does it take to select a target that is 21 inch away and 3 inch wide?

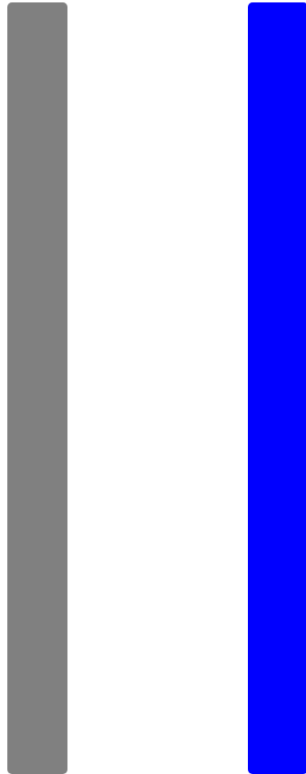
$$\begin{aligned} MT &= 0.028 + 0.112 * \log_2(1+7) \\ &= 0.028 + 0.112 * \log_2(8) \\ &= 0.028 + 0.112 * 3 \\ &= 0.364ms \end{aligned}$$

Device Characteristics (**a** and **b** parameters)



Fitts' Law Experiment

<http://ergo.human.cornell.edu/FittsLaw/FittsLaw.html>



When blue rectangle appears,
click on it as fast as possible

Error: 0
 $MT = 315.501 + 108.539 \log(A/W + 1)$
RSquare = 0.904

Mean MT

A	W	MT	ID
512	16	827	5
128	64	445	2
128	16	725	3
512	64	672	3

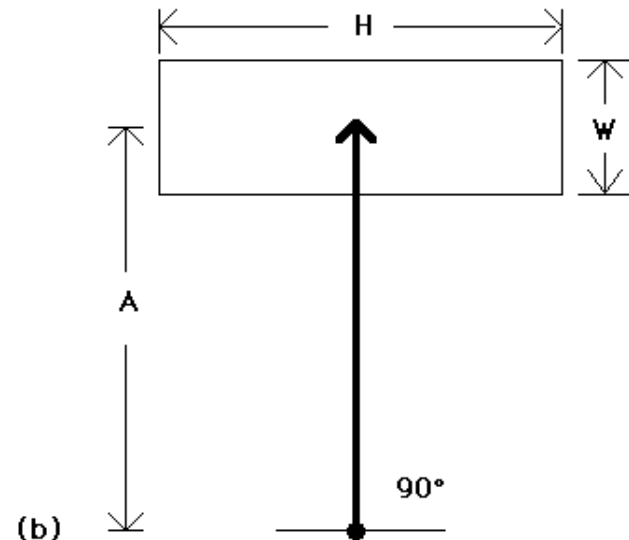
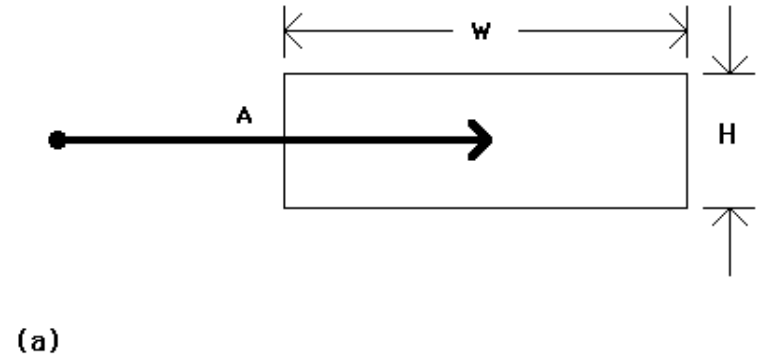
Raw Data

A	W	MT	ID
512	16	670	5
512	16	825	5
512	16	927	5
512	16	808	5
512	16	937	5
512	16	647	5
512	16	1145	5
512	16	856	5
512	16	631	5
128	64	543	2
128	64	376	2
128	64	456	2
128	64	504	2
128	64	409	2
128	64	415	2
128	64	353	2
128	64	519	2
128	64	433	2
128	16	624	3

Other example: <http://www.simonwallner.at/ext/fitts/>

2D Targets?

- If the approach angle is 0° (Figure a), then the 1D scenario applies.
- If the approach angle is 90° (Figure b), then the roles of width and height reverse
- Unfortunately, this has not been accommodated in past research.

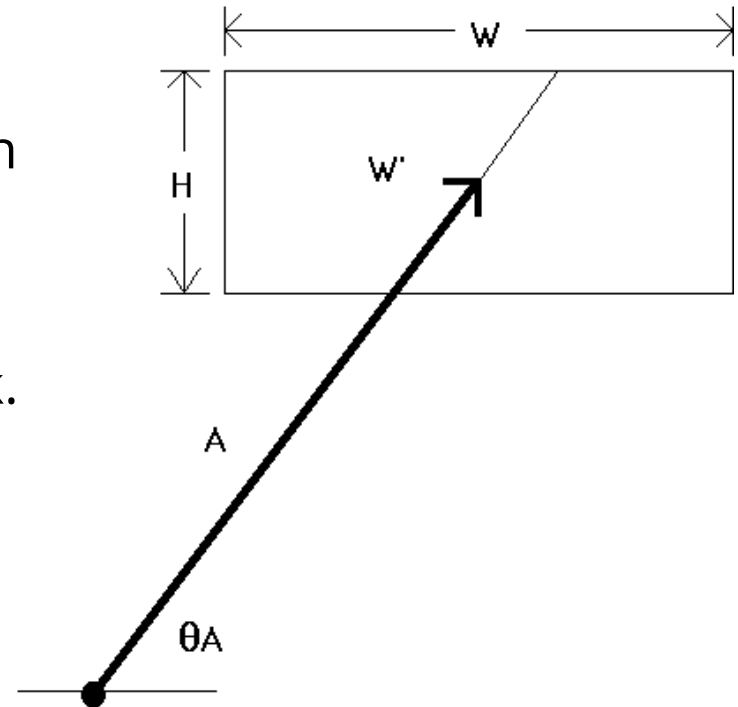


<http://www.yorku.ca/mack/CHI92.html>

(remember 'A' = Amplitude = 'D' = Distance)

2D Targets: W' as Cross Section Given Approach

- Measuring W along the approach axis.
- This is shown as W' (read "W prime") in the figure.
- W' model is appealing because it allows a 1D interpretation of a 2D task.
- But hard to know approach angle a priori ...



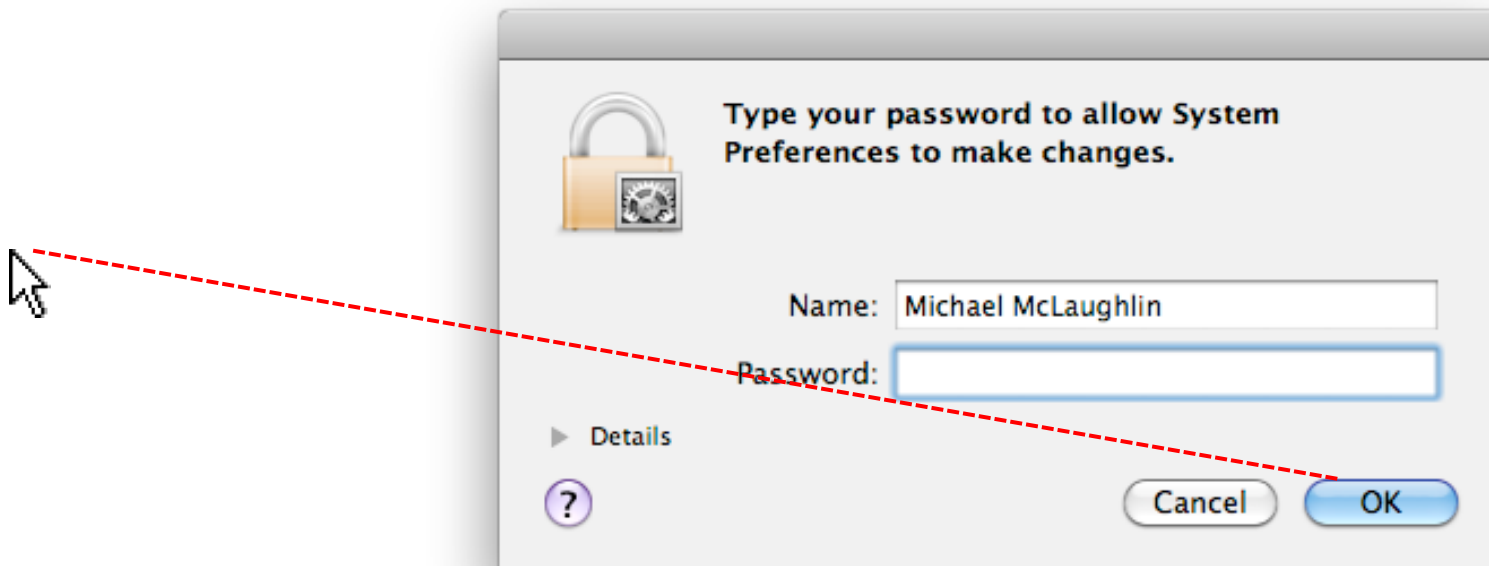
2D Targets: “W” is Minimum of Target W and H

$$MT = a + b \log_2 \left(\frac{D}{\min(W, H)} + 1 \right)$$

... but usually just write W assuming it's the minimum of target W and H

Fitts' Law Example

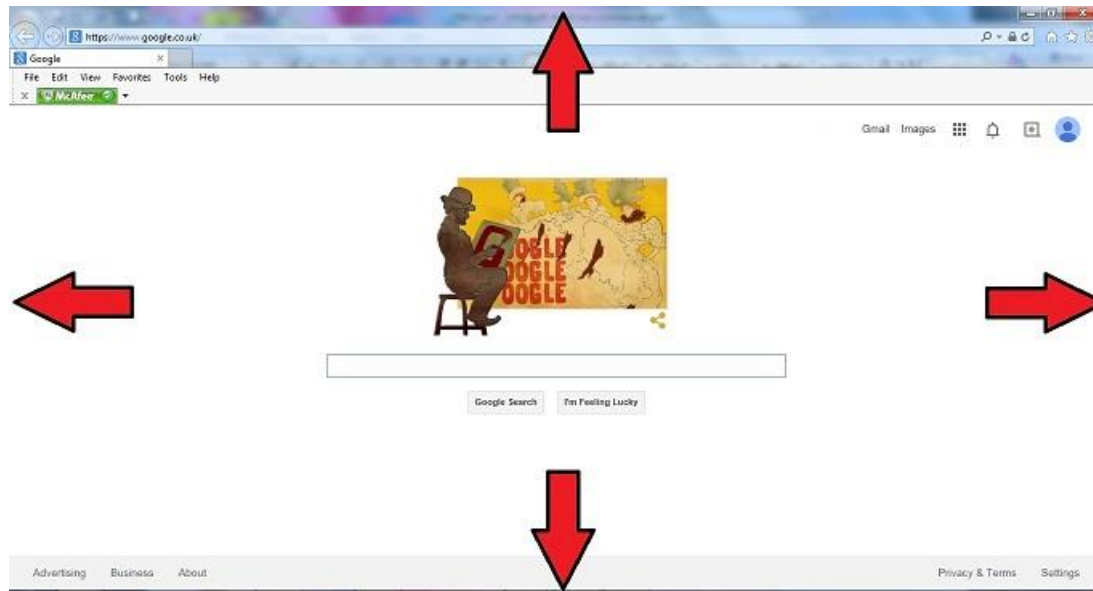
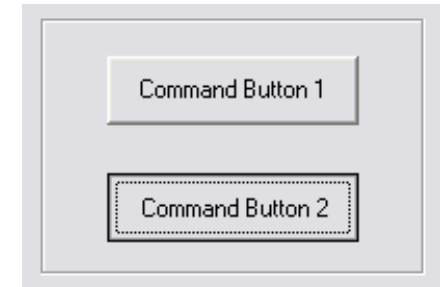
- Using a mouse to point ($a = -107$ and $b = 223$), what is the movement time to click on a 80 pixel by 32 pixel Cancel button located 400 pixels away?



$$MT = a + b \log_2 \left(\frac{D}{\min(W, H)} + 1 \right)$$

Applying Fitts's Law to User Interface Design

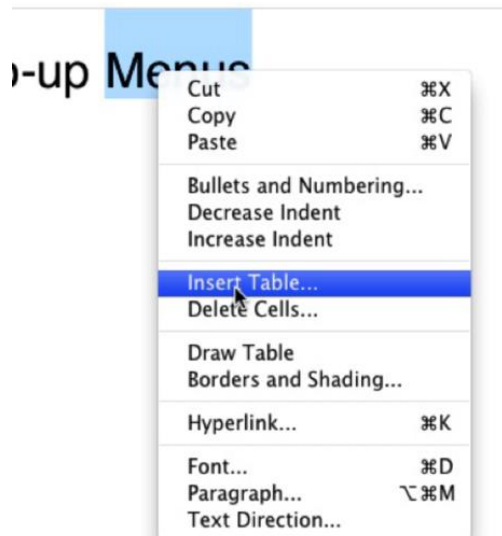
- Command buttons and any other interactive element in the graphical user interface must be distinguished from other non-interactive elements by size
- The outer edges and corners of the graphical user interface can be acquired with greater speed than anywhere else in the display, due to the pinning action of the screen.



Applying Fitts's Law to User Interface Design

- Pop-up menus better support immediate selection of interactive elements than dropdown menus as the user does not have to move the cursor from its current position.
- Context Menu lowers D, but some items closer than others
- Pie Menus near mouse, all items same D (optimal)

Pop-up/context menu

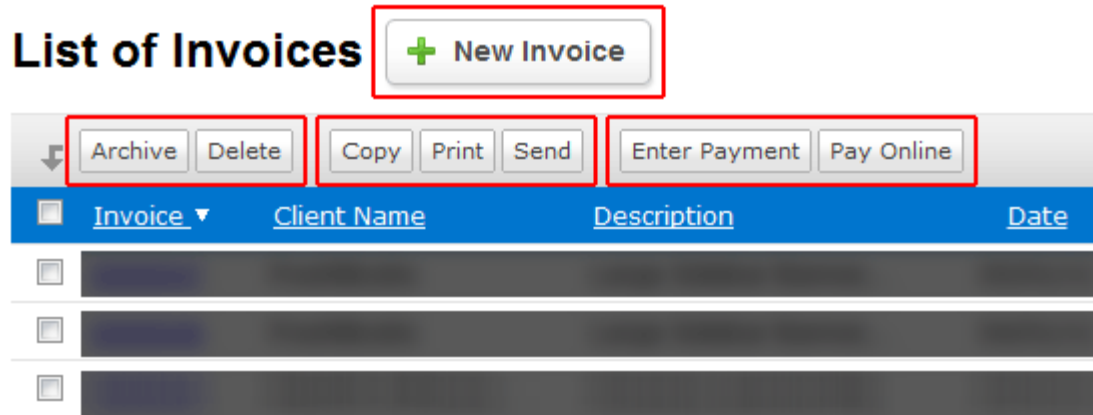


Pie menu



Applying Fitts's Law to User Interface Design

- Placing key site components far apart will increase the amount of time required in completing sequential tasks in your interface.



Fitts' Law in the Wild

- A field study of aimed movements in graphical user interfaces, designed to get better insight into pointing in the real world and assess the validity of Fitts' law in the wild
- Over 2 million aimed movements and about 1 billion pixels covered by the cursor
- Fitts' law is indeed robust for modeling pointing performance

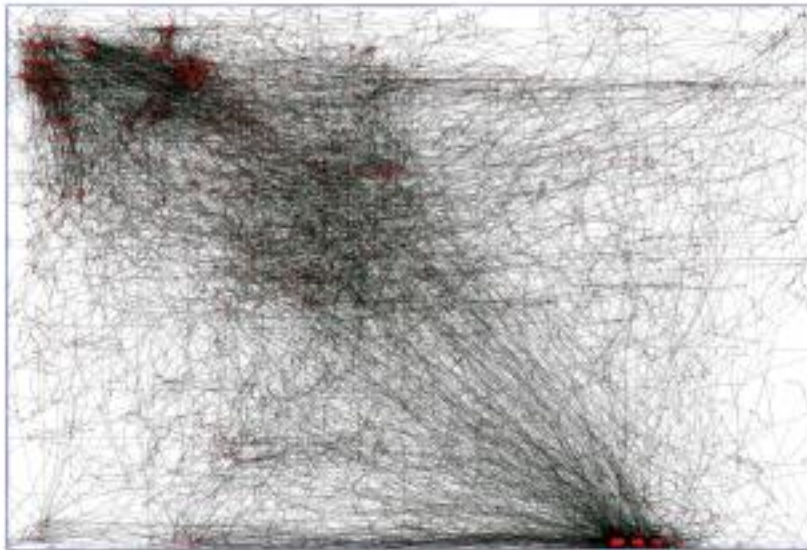


Figure 1. Mouse trajectories (black) and clicks (red).

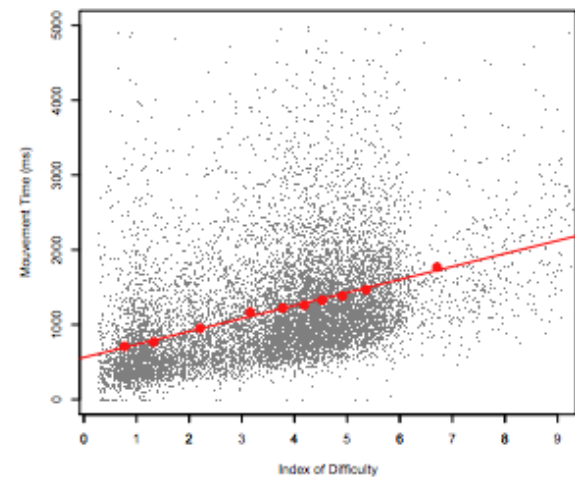
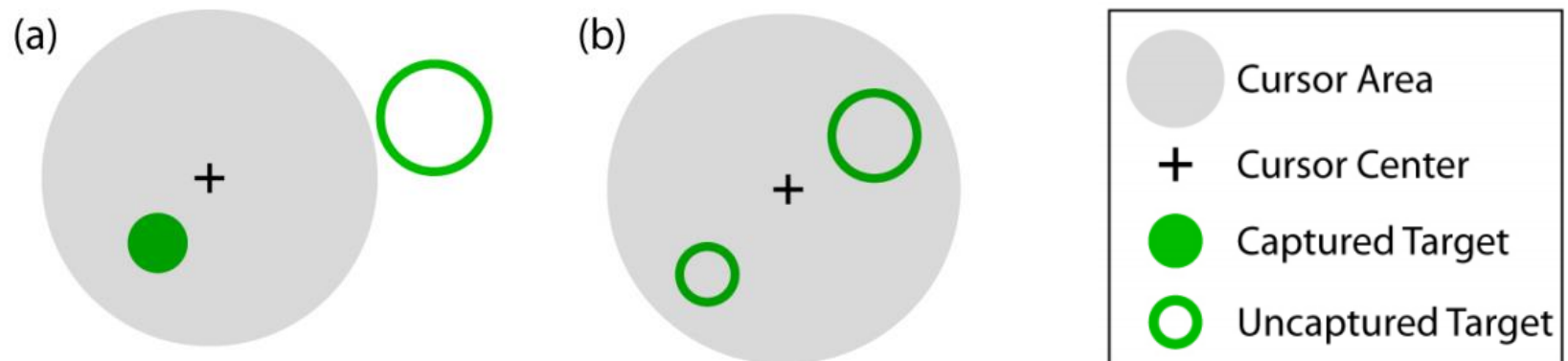


Figure 7. Fitts' linear fit of MT by ID using the averaging process for one user (# 4). Small (black) points represent all the points. Large (red) points are the means of each quantile.

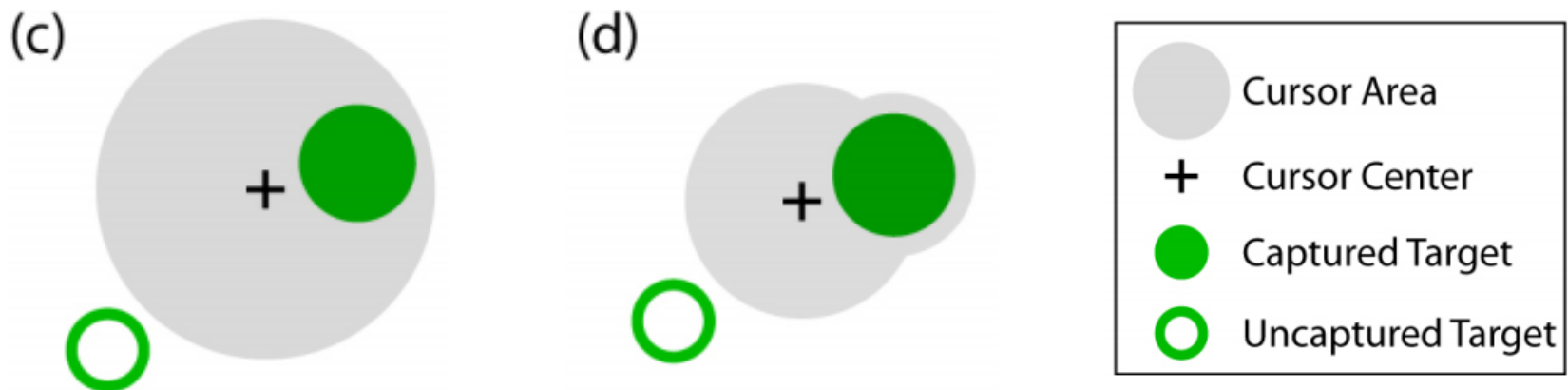
Area Cursor

- A standard point cursor commonly has a single point of activation or hotspot
- Area cursors have larger hotspots defined by the boundary of the cursor (Figure a)
- Problems arise, however, when the area cursor encompasses more than one target
- Making it difficult to isolate the intended target (Figure b)



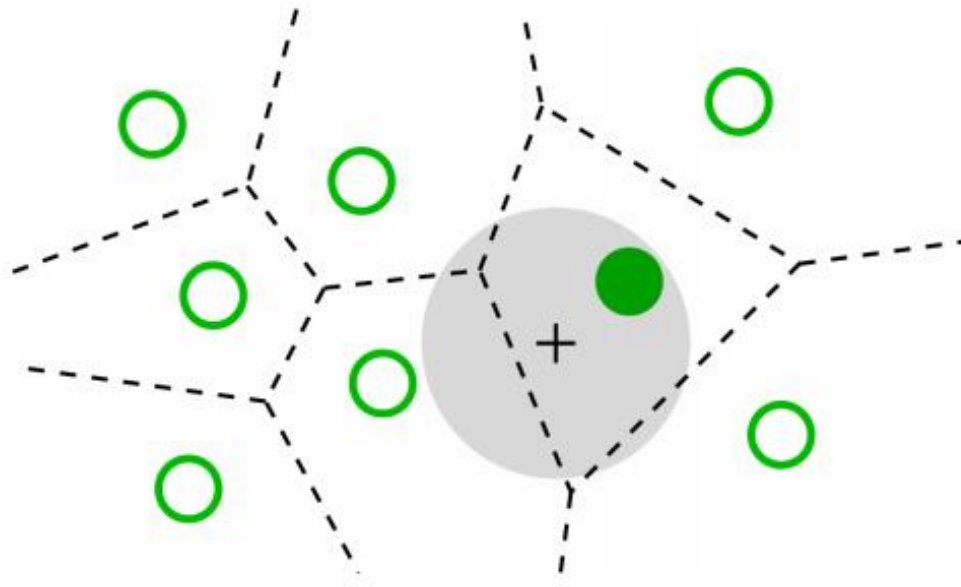
Bubble Cursor

- The bubble cursor solves this problem of the area cursor by dynamically updating its size based on the proximity of surrounding targets, such that there is always exactly one target inside the hotspot (Figure c, d).
- The bubble cursor morphs to encompass a target when the basic circular cursor cannot completely do so without intersecting a neighboring target (Figure d).

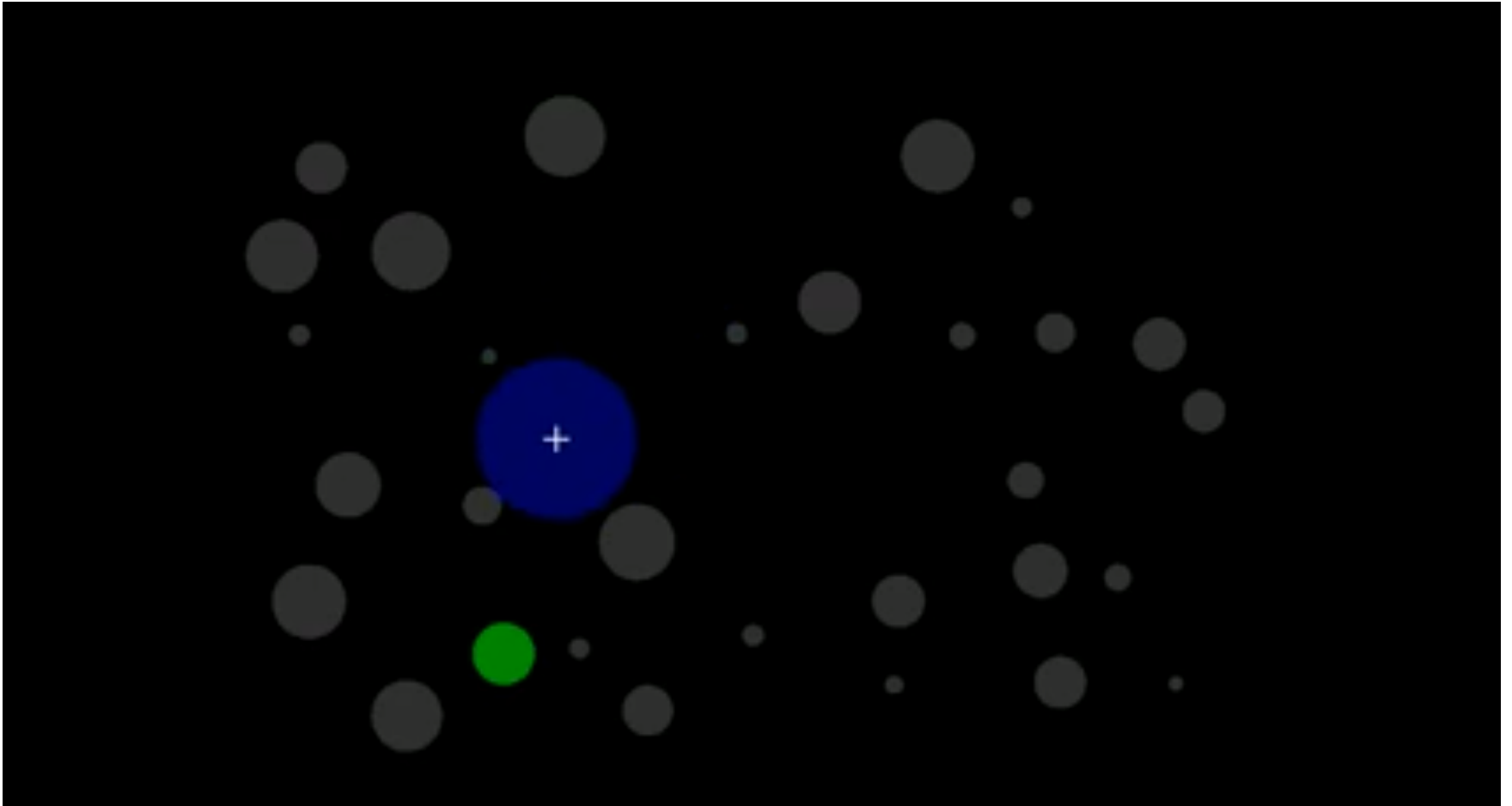


Bubble Cursor

- Fitts Law: decreasing D and increasing W
- Using the bubble cursor essentially divides up the total space in which all targets reside into regions
- There is exactly one target inside each region, and that target is the closest target to any point within that region. In mathematical terms, this is referred to as a **Voronoi diagram**

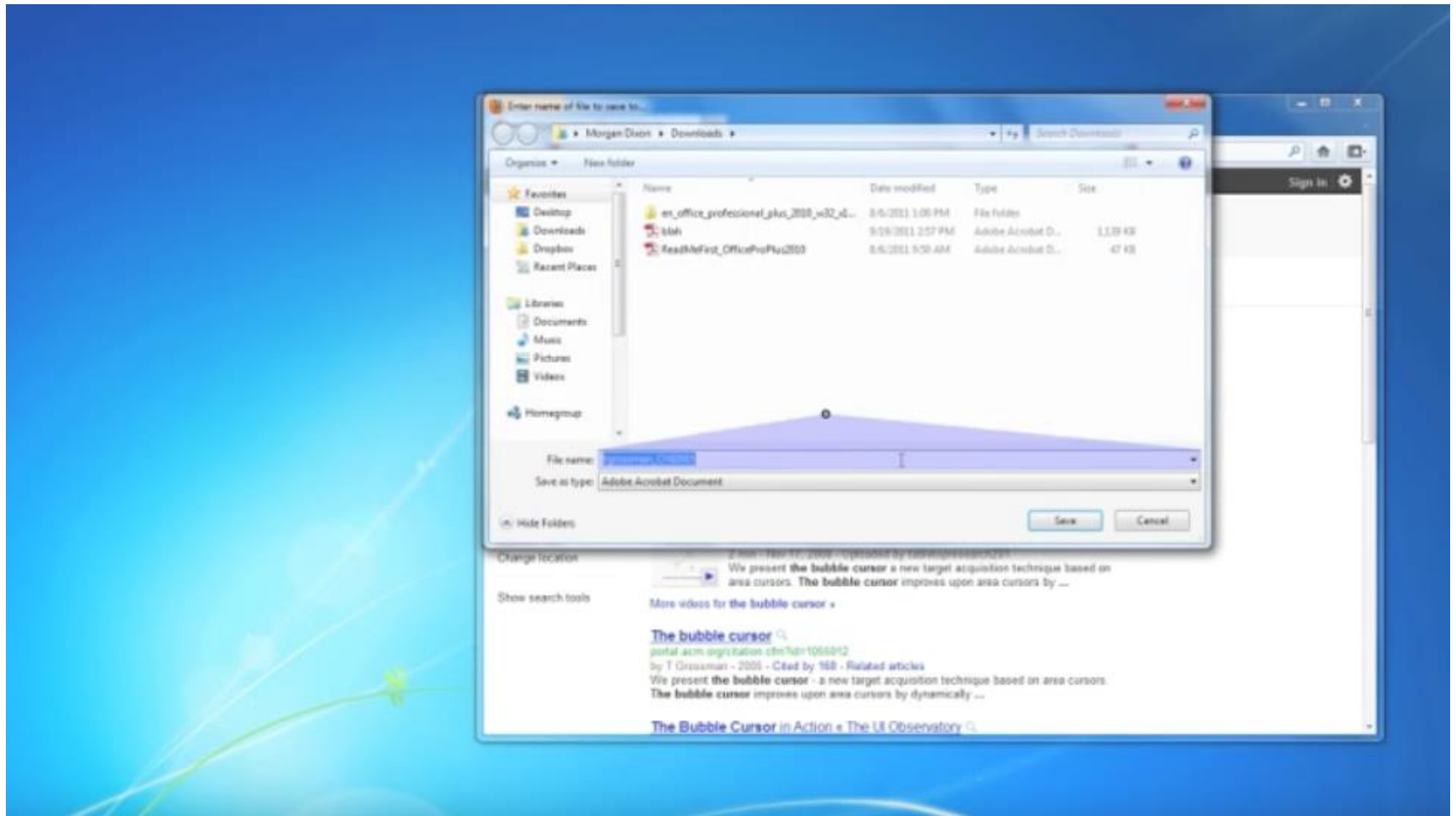


Bubble Cursor



Bubble Cursor (Grossman and Balakrishnan, 2005)

- http://youtu.be/JUBXkD_8ZeQ



A General-Purpose Bubble Cursor using Prefab (Dixon et al. 2012)

- https://youtu.be/46EopD_2K_4

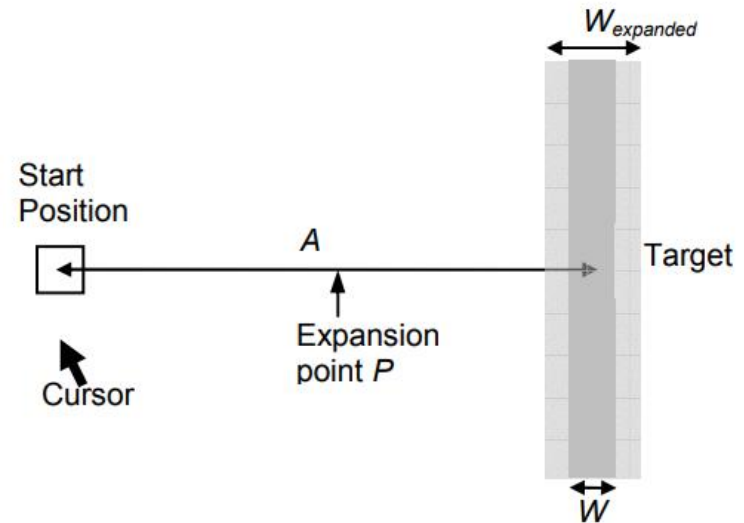
OSX Dock Expansion

- OSX Dock expands in visual space, but not motor space ...
- Fitts's law says selecting an expanded target on the dock is no easier than the default small targets



Acquisition of Expanding Targets

- The size of the interface widget or viewing region dynamically changes to provide the user with a larger target area
 - What happens when the target size changes after the user has already begun movement towards that target
-
- In the base case, the target had a width of W .
 - In the expanded cases, the target began with a width W but expanded to W_{expanded} when the cursor moved past a specified expansion point P .



Acquisition of Expanding Targets

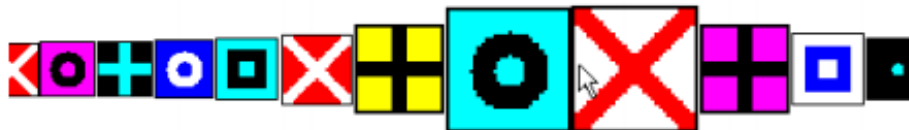
- Results indicate that such tasks can indeed be accurately modeled by Fitts' law.
- User performance is consistently aided by the target expansion, even when the target only expands after 90% of the distance towards the target has been traversed.



(a)



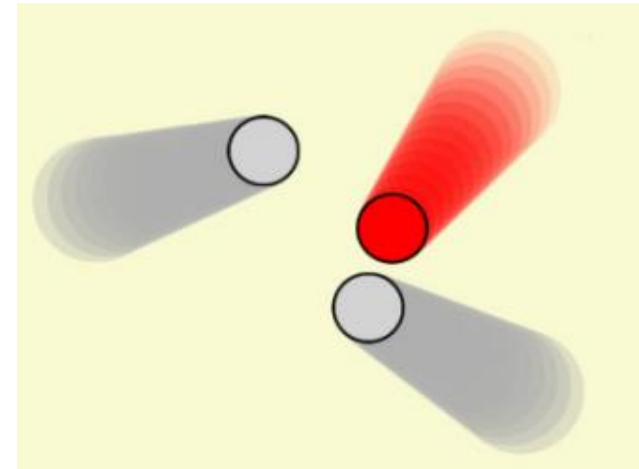
(b)



(c)

Comet and Target Ghost

- Comet, a technique that enhances targets based on their speed and direction
- With Comet the target's activation size is increased with an appended tail, analogous to comets that are seen in the sky
- Compared with bubble cursor for 1D (movement along x-axis) and 2D selection tasks
- In a 1D selection task, Comet outperformed the Area and Bubble cursors
- In a 2D selection task, Comet outperformed the Bubble, but only in conditions without the ghost



Khalad Hasan, Tovi Grossman, and Pourang Irani. 2011. Comet and target ghost: techniques for selecting moving targets. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '11). ACM, 839–848.

Motor Space vs. Screen Space

- Dynamically change CD Gain based on position of cursor
 - Making the cursor move more slowly when over the save button makes it larger in “motor space” even though it looks the same size in “screen space”.
 - LOOKS the same on screen, but “Save” button is “sticky”.
 - Faster to click “Save” (if Fitts’ Law calculated in motor space).

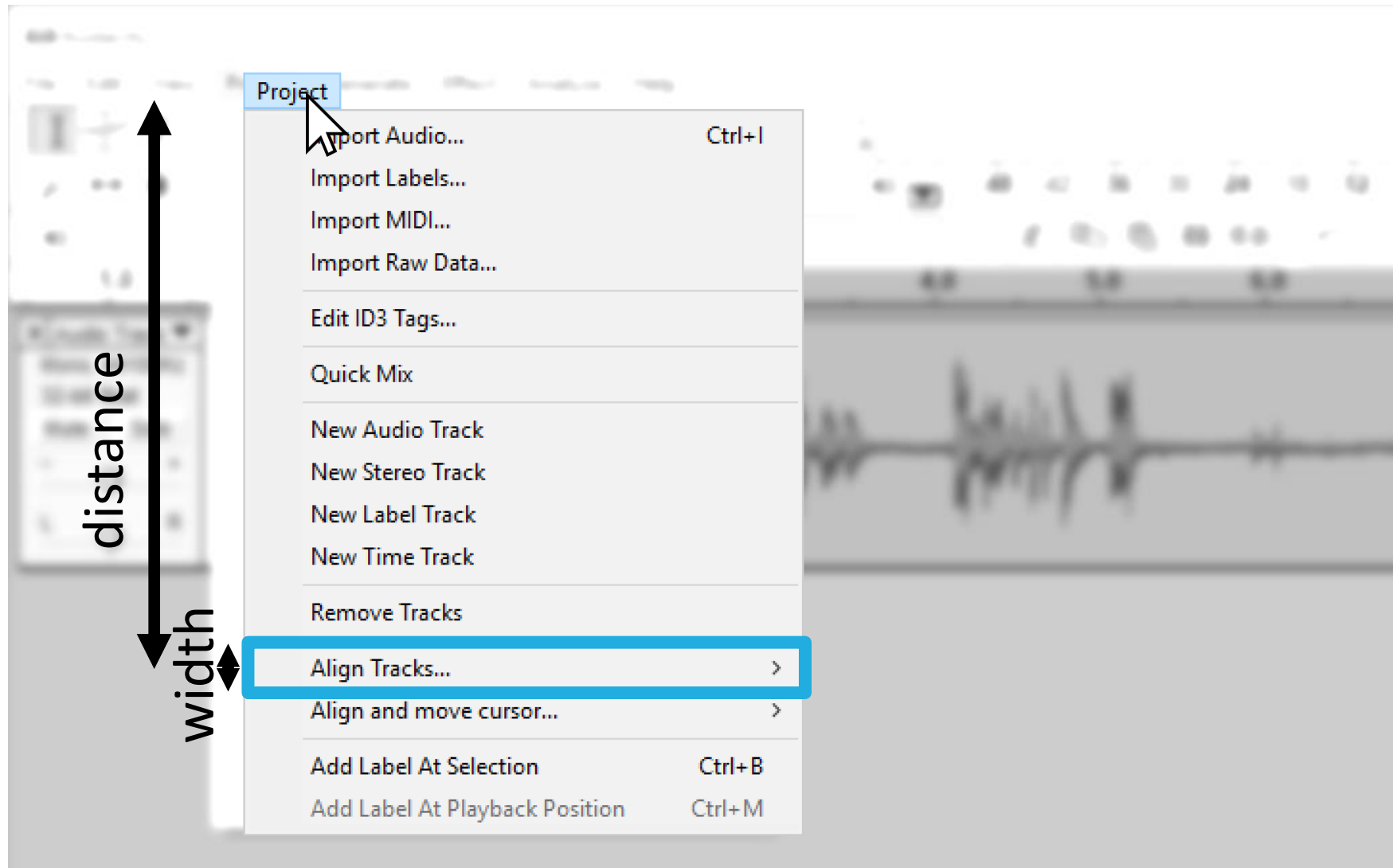


visual space

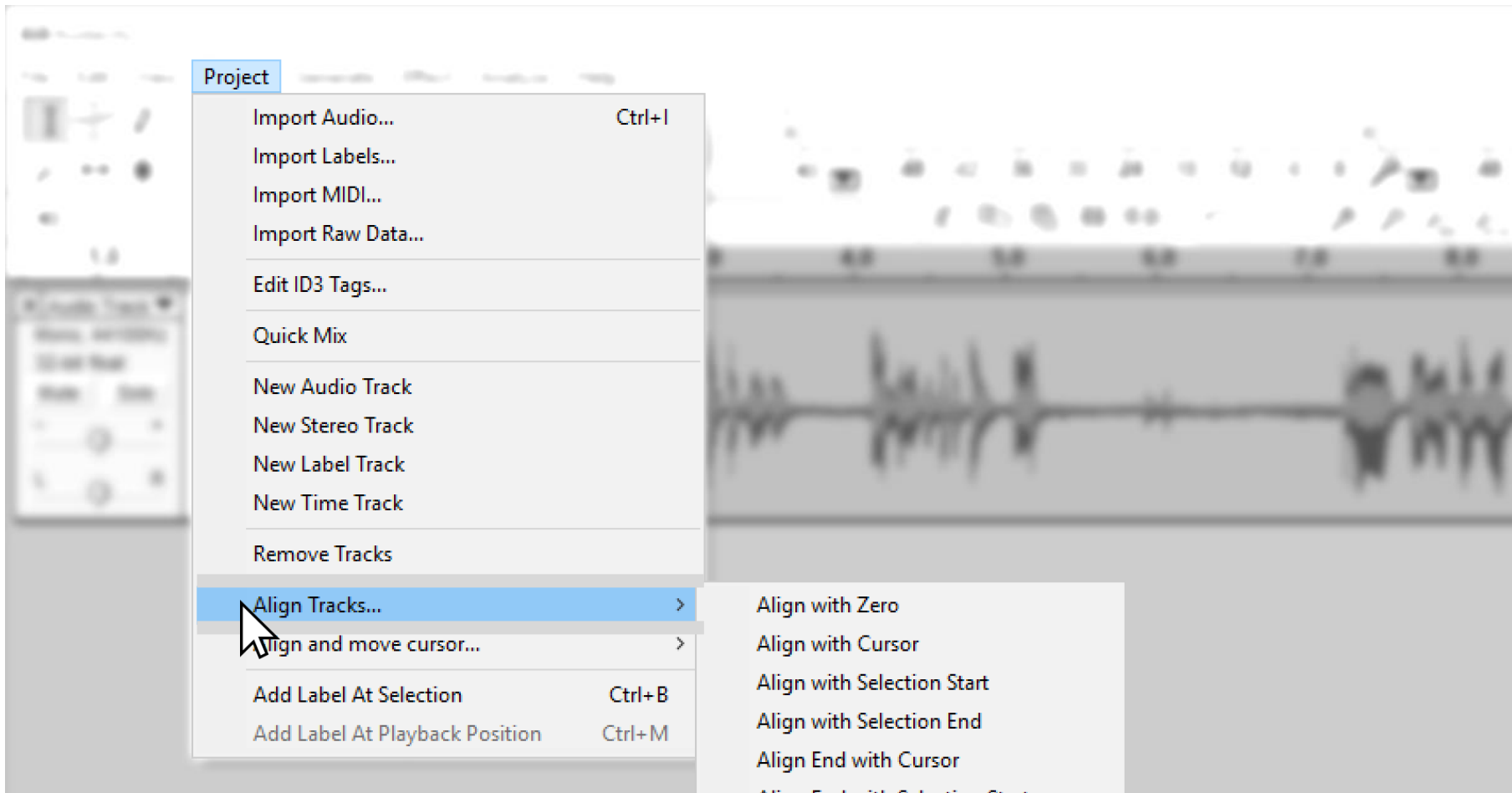


motor space

Selecting a Sub-Menu



Selecting a Sub-Menu



Steering Law

- Steering Law is an adaptation of Fitts' Law, developed by Zhai and Acott
- Choose a paradigm which focuses on steering between boundaries
- Applicability?

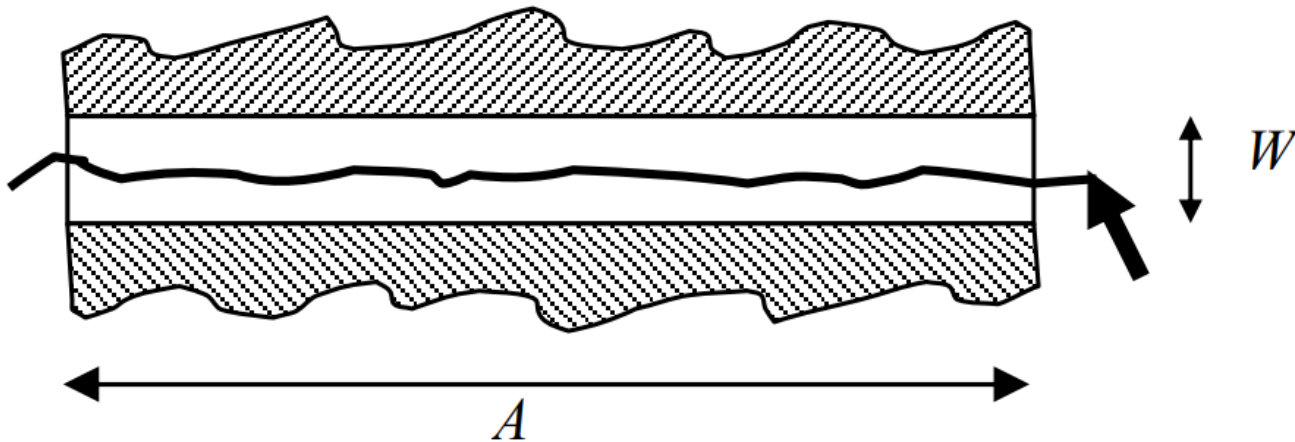
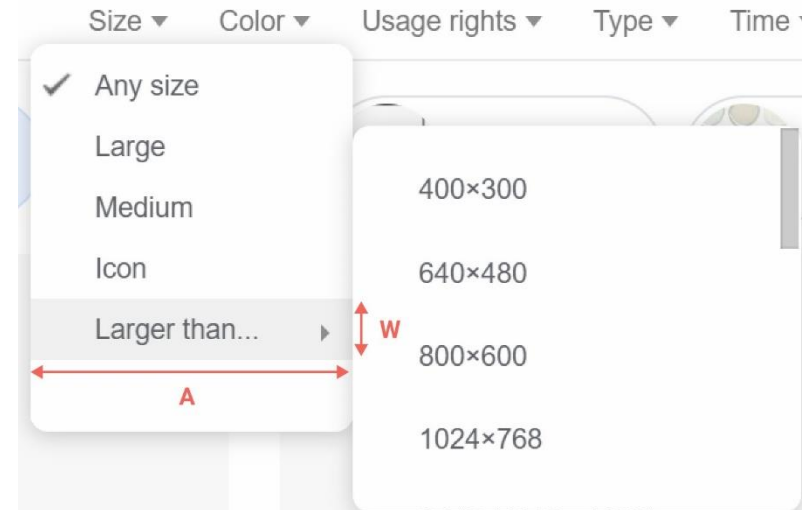
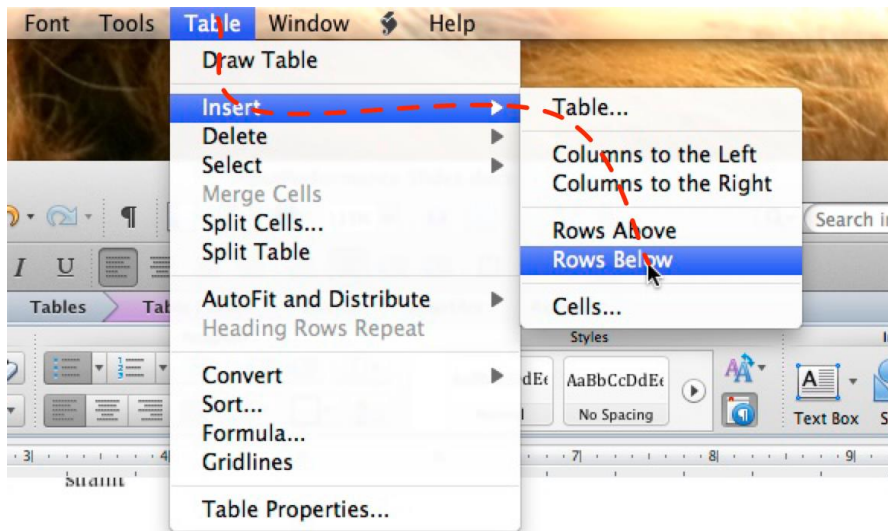


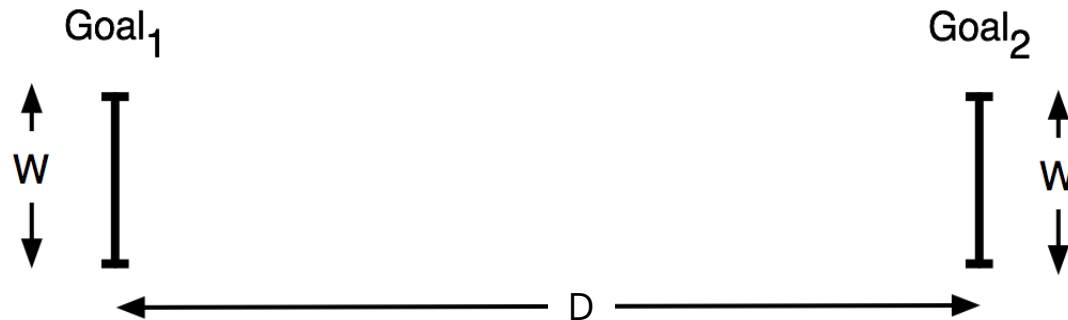
Figure 1: This movement through a tunnel constraint is modeled by the steering law

Steering Law

- The Steering Law predicts the time necessary to steer a pointer through a bounded tunnel (such as a menu, a scroll bar, or slider).
- The steering time depends on the length and the width of the tunnel
 - the longer and the narrower the tunnel, the more time will be required to successfully steer through it



Steering Law: Goal Passing



- Subjects passed a stylus from one end to the other
 - As fast as possible
 - Between each goal
 - Several trials with different amplitudes (D) and widths (W)
- Result: Same law as Fitts' tapping task

(Accot-Zhai) Steering Law

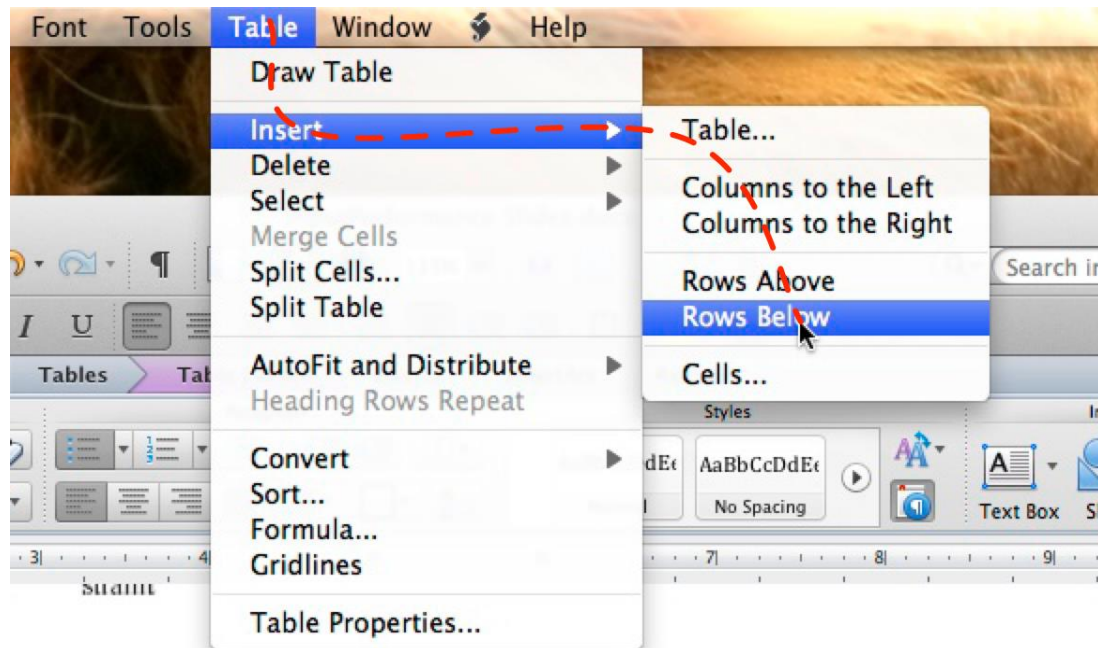
- Accot, J., & Zhai, S. (1997). Beyond Fitts' law: models for trajectory-based HCI tasks. CHI'97
- Tunnel traveling task is not related to the logarithm of D/W but to D/W

$$MT = a + b \frac{D}{W} \quad ID = \frac{D}{W}$$

- The movement time (MT) to acquire a target through a tunnel is a function of the length (D) and width (W) of the tunnel. It depends on the input device.
 - MT: movement time
 - a and b: constants dependent on the pointing system
 - D: distance, i.e. length of the tunnel
 - W: width of the tunnel

Hierarchical Menus

- Sum the parts of the path:
 - Wide path (but short stopping distance)
 - Narrow path (but wide stopping distance)
 - Wide path (with short stopping distance)



Summary

- We have mathematical models for acquiring a target, both when the path is unconstrained and constrained
 - Larger/closer is faster
- Gives some ideas for speeding things up
 - Keep things close (contextual, pie-menus)
 - Make things larger (bubble cursors)
 - Manipulate motor space to make intended targets stickier